COUNTING

⌖ PRODUCT RULE ***

1. A new company with just two employees, Sanchez and Patel, rents a floor of a building with 12 offices. How many ways are there to assign different offices to these two employees?
2. The chairs of an auditorium are to be labeled with an uppercase English letter followed by a positive integer not exceeding 100. What is the largest number of chairs that can be labeled differently?
3. There are 32 computers in a data center in the cloud. Each of these computers has 24 ports. How many different computer ports are there in this data center?
4. How many different bit strings of length seven are there?
5. How many different license plates can be made if each plate contains a sequence of three uppercase English letters followed by three digits (and no sequences of letters are prohibited, even if they are obscene)?
6. How many different three-letter initials can people have?
7. How many different three-letter initials with none of the letters repeated can people have?
8. How many different three-letter initials are there that begin with an *A*?
9. How many bit strings are there of length eight?
10. In how many ways can an organization containing 26 members elect a president, treasurer, and secretary (assuming no person is elected to more than one position)?
11. Suppose a license plate contains two letters followed by three digits with the first digit not zero. How many different license plates can be printed?

⌖ SUM RULE ***

1. Suppose that either a member of the mathematics faculty or a student who is a mathematics major is chosen as a representative to a university committee. How many different choices are there for this representative if there are 37 members of the mathematics faculty and 83 mathematics majors and no one is both a faculty member and a student?
2. A student can choose a computer project from one of three lists. The three lists contain 23, 15, and 19 possible projects, respectively. No project is on more than one list. How many possible projects are there to choose from?

⌖ More Complex Counting Problems ***

1. In a version of the computer language BASIC, the name of a variable is a string of one or two alphanumeric characters, where uppercase and lowercase letters are not distinguished. (An *alphanumeric* character is either one of the 26 English letters or one of the 10 digits.) Moreover, a variable name must begin with a letter and must be different from the five strings of two characters that are reserved for programming use.
   How many different variable names are there in this version of BASIC?
2. How many bit strings are there of length six or less, not counting the empty string?
3. Each user on a computer system has a password, which is six to eight characters long, where each character is an uppercase letter or a digit. Each password must contain at least one digit.
   How many possible passwords are there?
4. There are 18 mathematics majors and 325 computer science majors at a college.
   a) In how many ways can two representatives be picked so that one is a mathematics major and the other is a computer science major?
   b) In how many ways can one representative be picked who is either a mathematics major or a computer science major?
5. A multiple-choice test contains 10 questions. There are four possible answers for each question.
   a) In how many ways can a student answer the questions on the test if the student answers every question?

b) In how many ways can a student answer the questions on the test if the student can leave answers blank?

6. A particular brand of shirt comes in 12 colors, has a male version and a female version, and comes in three sizes for each sex. How many different types of this shirt are made?

7. How many 5-element DNA sequences

a) end with A?

b) start with T and end with G?

c) contain only A and T?

d) do not contain C?

8. How many 6-element RNA sequences

a) do not contain U?

b) end with GU?

c) start with C?

d) contain only A or U?

9. How many license plates can be made using either two uppercase English letters followed by four digits or two digits followed by four uppercase English letters?

10. How many license plates can be made using either three uppercase English letters followed by three digits or four uppercase English letters followed by two digits?

11. How many license plates can be made using either two or three uppercase English letters followed by either two or three digits?

12. Suppose a license plate containing three letters followed by two digits with the last digit not zero. How many different license plates can be printed? **(AU22)**

⌗ The Subtraction Rule **

1. A computer company receives 350 applications from college graduates for a job planning a line of new web servers. Suppose that 220 of these applicants majored in computer science, 147 majored in business, and 51 majored both in computer science and in business. How many of these applicants majored neither in computer science nor in business?

2. How many bit strings of length eight either start with a 1 bit or end with the two bits 00?

3. How many bit strings of length ten both begin and end with a 1?

4. How many positive integers not exceeding 1000 are divisible by 7 or 11?

⌗ Division

1. How many positive integers between 5 and 31 **(AU23)**

a) are divisible by 3? Which integers are these?

b) are divisible by 4? Which integers are these?

c) are divisible by 3 and by 4? Which integers are these?

2. How many positive integers between 50 and 100

a) are divisible by 7? Which integers are these?

b) are divisible by 11? Which integers are these?

c) are divisible by both 7 and 11? Which integers are these?

3. How many positive integers less than 1000

a) are divisible by 7?

b) are divisible by 7 but not by 11?

c) are divisible by both 7 and 11?

d) are divisible by either 7 or 11?

e) are divisible by exactly one of 7 and 11?

f) are divisible by neither 7 nor 11?

g) have distinct digits?

h) have distinct digits and are even?

4. How many positive integers between 100 and 999 inclusive

a) are divisible by 7?

b) are odd?

c) have the same three decimal digits?

d) are not divisible by 4?

e) are divisible by 3 or 4?
f) are not divisible by either 3 or 4?
g) are divisible by 3 but not by 4?
h) are divisible by 3 and 4?

5. How many positive integers between 1000 and 9999 inclusive
a) are divisible by 9?
b) are even?
c) have distinct digits?
d) are not divisible by 3?
e) are divisible by 5 or 7?
f) are not divisible by either 5 or 7?
g) are divisible by 5 but not by 7?
h) are divisible by 5 and 7?

⌖ The Pigeonhole Principle ***

1. Among any group of 367 people, there must be at least two with the same birthday, because there are only 366 possible birthdays.

2. In any group of 27 English words, there must be at least two that begin with the same letter, because there are 26 letters in the English alphabet.

3. How many students must be in a class to guarantee that at least two students receive the same score on the final exam, if the exam is graded on a scale from 0 to 100 points?

4. What is the minimum number of students required in a discrete mathematics class to be sure that at least six will receive the same grade, if there are five possible grades, A, B, C, D, and F?

5. What is the least number of area codes needed to guarantee that the 25 million phones in a state can be assigned distinct 10-digit telephone numbers? (Assume that telephone numbers are of the form *NXX-NXX-XXXX*, where the first three digits form the area code, *N* represents a digit from 2 to 9 inclusive, and *X* represents any digit.)

6. A bowl contains 10 red balls and 10 blue balls. A woman selects balls at random without looking at them.
a) How many balls must she select to be sure of having at least three balls of the same color?
b) How many balls must she select to be sure of having at least three blue balls?

7. State the *pigeonhole principle*. Use pigeon hole principle to show that among any group of **20** people (where any two people are either friends or enemies), there are either **four** mutual friends or **four** mutual enemies. **(AU23)**

8. State the generalized pigeonhole principle. What is the minimum of students required in a Discrete Mathematics class to be sure that at least eight will receive the same grade, if there are four possible grades, namely A+, A, Aand F**. (SP23)**

9. In your dresser drawer you have a jumble of socks in twocolors, say blue and gray. It’s dark, and you don’t want to wake your spouse. **(SP23)**
i. How many socks must you grab to guarantee that you have a pair of the same color?
ii. How many socks must you grab to guarantee that you have 3 socks of the same color?

10. You have pens of many colors. In the drawer there are 43 black pens,2 red pens,23 blue pens and 8 orange pens. What is the minimum numbers of pens you need to take off from the drawer in order to be certain that you have taken four pens of same colors?

11. Explain how pigeonhole principle can be used to show that among any 11 integers, at least two must have the same last digit. **(AU22)**

⌖ Permutations ***

1. In how many ways can we select three students from a group of five students to stand in line for a picture? In how many ways can we arrange all five of these students in a line for a picture?

2. Let $S = \{1, 2, 3\}$. The ordered arrangement 3, 1, 2 is a permutation of *S*. The ordered arrangement 3, 2 is a 2-permutation of *S*.

3. How many ways are there to select a first-prize winner, a second-prize winner, and a third-prize winner from 100 different people who have entered a contest?
4. Suppose that there are eight runners in a race. The winner receives a gold medal, the second place finisher receives a silver medal, and the third-place finisher receives a bronze medal. How many different ways are there to award these medals, if all possible outcomes of the race can occur and there are no ties?
5. Suppose that a saleswoman has to visit eight different cities. She must begin her trip in a specified city, but she can visit the other seven cities in any order she wishes. How many possible orders can the saleswoman use when visiting these cities?
6. How many different ways are there to select 4 different players from 10 players on a team to play four tennis matches, where the matches are ordered?
7. How many permutations of the letters *ABCDEFGH* contain the string *ABC* ?
8. How many permutations of the letters *ABCDEFG* contain
   a) the string *BCD*? **b)** the string *CFGA*? **c)** the strings *BA* and *GF*?
   d) the strings *ABC* and *DE*? **e)** the strings *ABC* and *CDE*? **f)** the strings *CBA* and *BED?*

⌖ Permutations with Indistinguishable Objects

1. How many seven-letter words can be formed using the word "BENZENE"?
2. How many different strings can be made by reordering the letters of the word SUCCESS.?
3. How many strings with five or more characters can be formed from the letters in *SEERESS*?
4. How many strings with seven or more characters can be formed from the letters in *EVERGREEN*?
5. How many ways are there to order the letters of the word *INDISCREETNESS*?

⌖ Permutations with Reptation

1. How many strings of length r can be formed from the uppercase letters of the English alphabet?

⌖ Combinations ***

1. How many different committees of three students can be formed from a group of four students?
2. Let *S* be the set {1, 2, 3, 4}. Then {1, 3, 4} is a 3-combination from *S*. (Note that {4, 1, 3} is the same 3-combination as {1, 3, 4}, because the order in which the elements of a set are listed does not matter.)
3. We see that $C(4, 2) = 6$, because the 2-combinations of $\{a, b, c, d\}$ are the six subsets $\{a, b\}$, $\{a, c\}$, $\{a, d\}$, $\{b, c\}$, $\{b, d\}$, and $\{c, d\}$.
4. Find the number of combinations of the four objects a,b,c,d, taken three at a time.?
5. How many poker hands of five cards can be dealt from a standard deck of 52 cards? Also, how many ways are there to select 47 cards from a standard deck of 52 cards?
6. How many ways are there to select five players from a 10-member tennis team to make a trip to a match at another school?
7. A group of 30 people have been trained as astronauts to go on the first mission to Mars. How many ways are there to select a crew of six people to go on this mission (assuming that all crew members have the same job)?
8. How many committees of 3 can be formed 8 people?
9. Suppose that there are 9 faculty members in the mathematics department and 11 in the computer science department. How many ways are there to select a committee to develop a discrete mathematics course at a school if the committee is to consist of three faculty members from the mathematics department and four from the computer science department?
10. How many bit strings of length 10 contain***
    a) exactly four 1s? **b)** at most four 1s?
    c) at least four 1s? **d)** an equal number of 0s and 1s?

11. How many bit strings of length 12 contain***

a) exactly three 1s?
b) at most three 1s?
c) at least three 1s?
d) an equal number of 0s and 1s?

12. A coin is flipped 10 times where each flip comes up either heads or tails. How many possible outcomes***

a) are there in total?
b) contain exactly two heads?
c) contain at most three tails?
d) contain the same number of heads and tails?

13. How many bit strings of length 10 have

a) exactly three 0s?
b) more 0s than 1s?
c) at least seven 1s?
d) at least three 1s?

14. How many strings of four decimal digits

a) do not contain the same digit twice?
b) end with an even digit?
c) have exactly three digits that are 9s?

15. How many strings of three decimal digits

a) do not contain the same digit three times?
b) begin with an odd digit?
c) have exactly two digits that are 4s?

Combinations with Reptation

1. How many ways are there to select five bills from a box containing at least five of each of the following denominations: $1, $2, $5, $10, $20, $50, and $100?
2. Suppose that a cookie shop has four different kinds of cookies. How many different ways can six cookies be chosen? Assume that only the type of cookie, and not the individual cookies or the order in which they are chosen, matters.
3. How many ways are there to select four pieces of fruit from a bowl containing apples, oranges, and pears if the order in which the pieces are selected does not matter, only the type of fruit and not the individual piece matters, and there are at least four pieces of each type of fruit in the bowl?

Binomial Coefficients and Identities ***

1. What is the coefficient of x12y13 in the expansion of $(x + y)^{25}$?
2. What is the coefficient of $x^{12}y^{13}$ in the expansion of $(2x - 3y)^{25?}$
3. What is the expansion of $(x + y)^4$?
4. Find the coefficient of x^5y^8 in $(x + y)^{13}$?
5. What is the coefficient of x^7 in $(1 + x)^{11}$?

NUMBER THEORY

Modular Arithmetic

1. Determine whether 17 is congruent to 5 modulo 6 and whether 24 and 14 are congruent modulo 6.
2. Find the value of $(19^3 \textbf{ mod } 31)^4 \textbf{ mod } 23$
3. Use the definition of addition and multiplication in $\mathbf{Z}_m$ to find $7 +_{11} 9$ and $7 \cdot_{11} 9$.
4. Find each of these values. ***(AU22)***
   a) (177 **mod** 31 + 270 **mod** 31) **mod** 31
   b) (177 **mod** 31 · 270 **mod** 31) **mod** 31

Primes

1. The prime factorizations of 100, 641, 999, and 1024?
2. Show that 101 is prime?
3. Find the prime factorization of 7007?
4. Find the prime factorization of 118724? **(SP23)**

Greatest Common Divisors and Least Common Multiples

1. Write down the differences between GCD and LCM? The answer should include appropriate example for both types? ***(AU22)***
2. What is the greatest common divisor of 24 and 36?
3. What is the greatest common divisor of 17 and 22?
4. Determine whether the integers 10, 17, and 21 are pairwise relatively prime and whether the integers 10, 19, and 24 are pairwise relatively prime.
5. Because the prime factorizations of 120 and 500 are $120 = 2^3 \cdot 3 \cdot 5$ and $500 = 2^2 \cdot 5^3$, find the greatest common divisor is
6. What is the least common multiple of $2^3 3^5 7^2$ and $2^4 3^3$?
7. Find the highest common factor of 34, 42 and 58

The Euclidean Algorithm

1. Find **GCD(1000, 625)** and **LCM(1000, 625)** using *Euclidean algorithm.* ***(AU23)***
2. Find the greatest common divisor of 414 and 662 using the Euclidean algorithm.
3. Express gcd(252, 198) = 18 as a linear combination of 252 and 198 by working backwards through the steps of the Euclidean algorithm.
4. Express gcd(252, 198) = 18 as a linear combination of 252 and 198 using the extended Euclidean algorithm. (SP23)

Linear Congruences

1. Find an inverse of 3 modulo 7 by first finding Bezout coefficients of 3 and 7.
2. Find an inverse of 101 modulo 4620.
3. What are the solutions of the linear congruence $3x \equiv 4 \pmod 7$?
4. Find all incongruent solutions to the following ***(SP23)***
   a. $7x \equiv 3 \pmod{15}$
   b. $x^2 \equiv 1 \pmod 8$
5. Define Linear Congruence. What are the solutions of the linear congruence $3x \equiv 4 \pmod 7$ **(AU23)**

The Chinese Remainder Theorem***

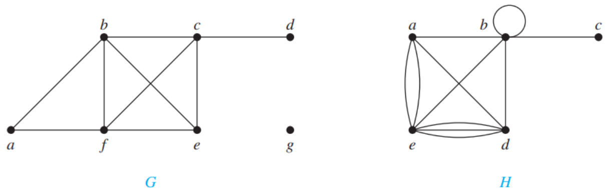
1. What are the solutions of the systems of congruences
   $x \equiv 2 \pmod 3$
   $x \equiv 3 \pmod 5$
   $x \equiv 2 \pmod 7$
2. Use the method of back substitution to find all integers x such that $x \equiv 1 \pmod 5$, $x \equiv 2 \pmod 6$, and $x \equiv 3 \pmod 7$.

3. Find the value of **x** using the *Chinese remainder theorem* ***(AU23)***
   $x \equiv 2 \pmod{3}$
   $x \equiv 1 \pmod{4}$
   $x \equiv 3 \pmod{5}$
4. Find value of x using Chinese remainder theorem **(SP23)**
   $x \equiv 1 \pmod{2}$
   $x \equiv 2 \pmod{3}$
   $x \equiv 3 \pmod{5}$
5. Find x using Chinese remainder theorem, if possible, such that **(AU22)**
   $2x \equiv 5 \pmod{7}$, and
   $3x \equiv 4 \pmod{8}$

GRAPH

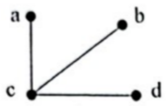
Graph Terminology

1. What are the degrees and what are the neighborhoods of the vertices in the graphs G and H displayed in Figure 1?

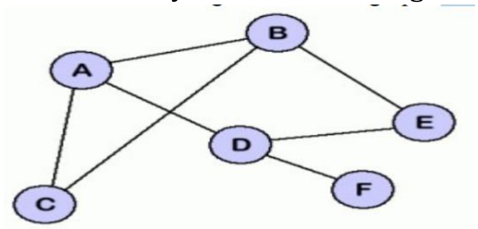


THE HANDSHAKING THEOREM

1. How many edges are there in a graph with 10 vertices each of degree six?
2. State the *Handshaking theorem*. Verify the Handshaking theorem for the following figure **(AU23)**

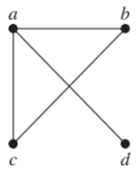


3. State the *Handshaking theorem*. Verify the Handshaking theorem for the following figure **(AU22)**



Representing Graphs

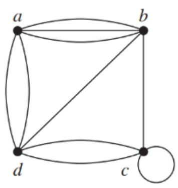
1. Use an adjacency matrix to represent the graph shown



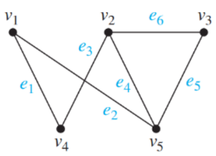
2. Draw a graph with the adjacency matrix

$$\begin{bmatrix} 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \end{bmatrix}$$

3. Use an adjacency matrix to represent the pseudograph shown



4. Represent the graph shown with an incidence matrix

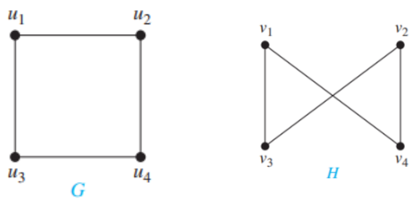


5. Draw the graph by the following Incidence Matrices (AU22)

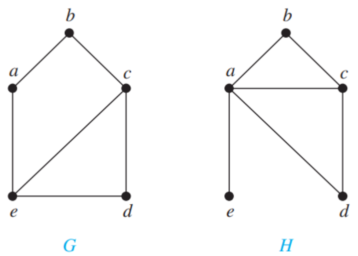
$$\begin{bmatrix} 1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 & 1 \\ 1 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 & 1 & 0 \end{bmatrix}$$

Isomorphism of Graphs

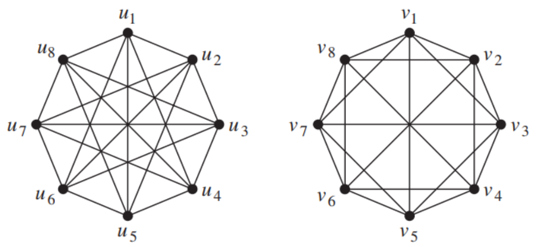
1. Show that the graphs $G = (V, E)$ and $H = (W, F)$, displayed in Figure 8, are isomorphic.



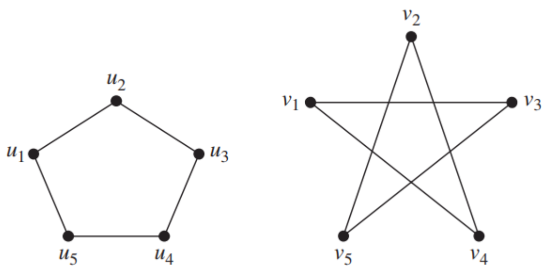
2. Show that the graphs displayed in Figure 9 are not isomorphic



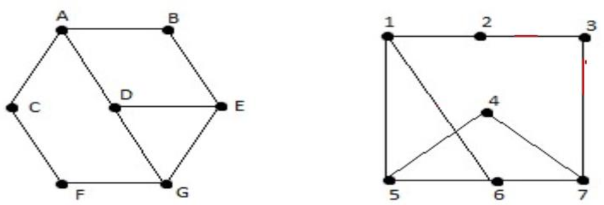
3. Are the graphs *isomorphic* or not? Please justify your answer **(AU23)**



4. Determine whether the following two graphs are isomorphic or not. **(SP23)**



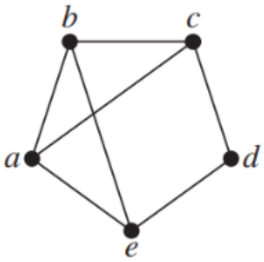
5. Determine whether the following two graphs are isomorphic or not. **(AU22)**



Euler Paths and Circuits

Determine whether the given graph has an Euler circuit. Construct such a circuit when one exists. If no Euler circuit exists, determine whether the graph has an Euler path and construct such a path if one exists.

1.



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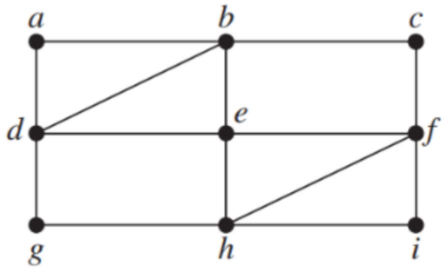

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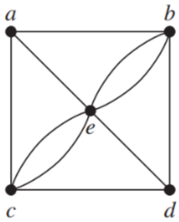

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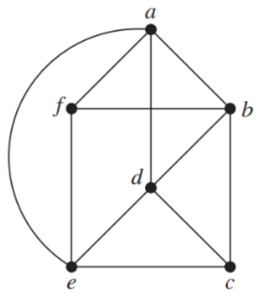

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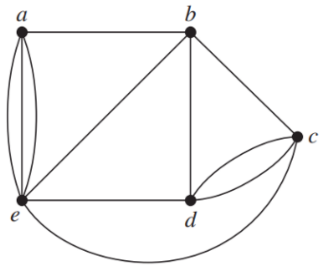

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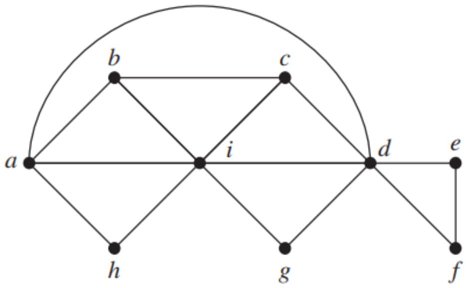

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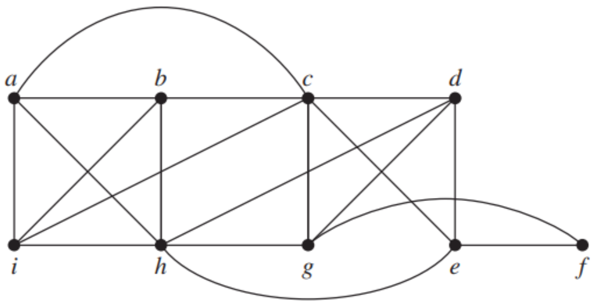

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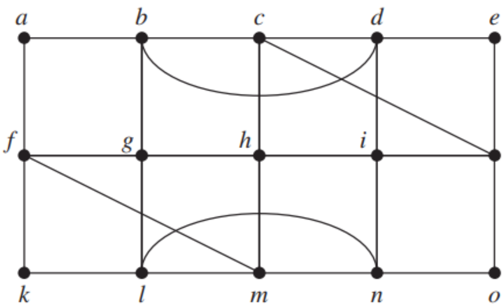

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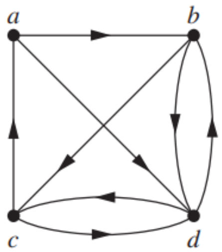

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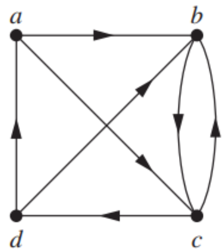

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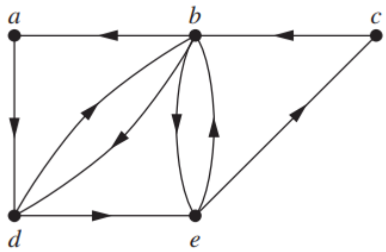

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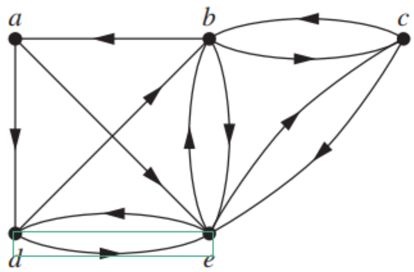

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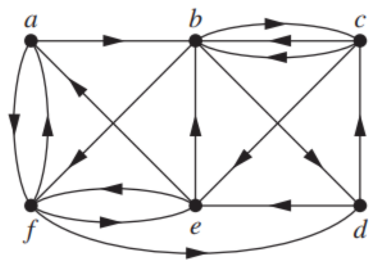
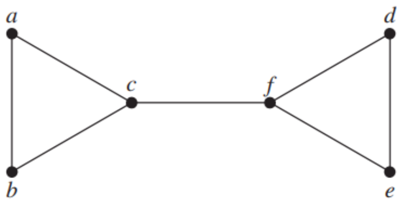


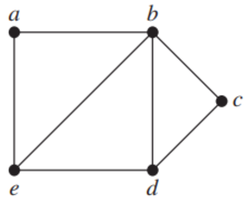
Hamilton Paths and Circuits

Determine whether the given graph has a Hamilton circuit. If it does, find such a circuit. If it does not, give an argument to show why no such circuit exists.

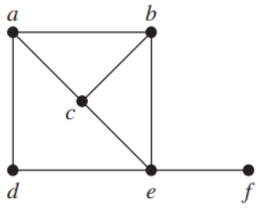
30.



31.



32.



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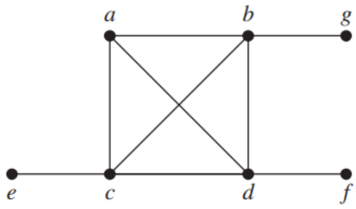

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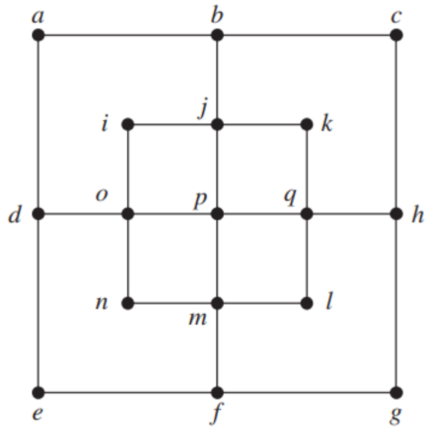

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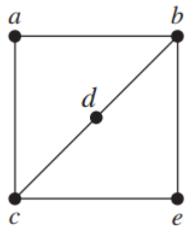

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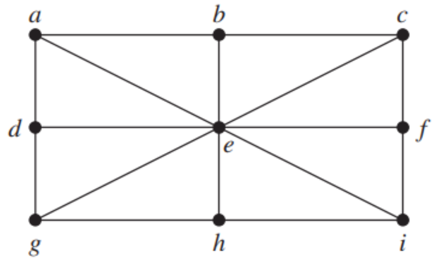
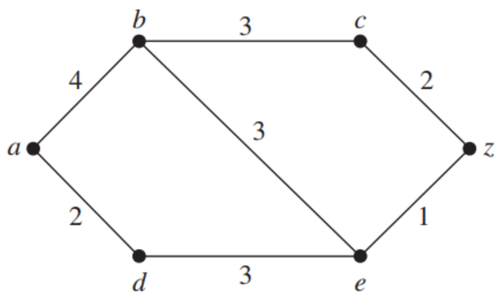
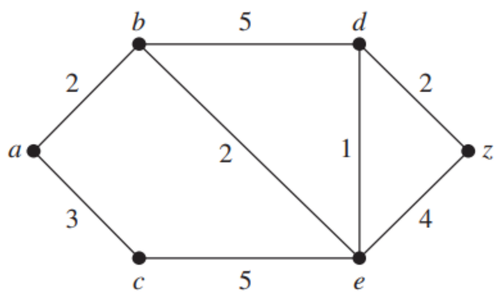


Shortest-Path Problems

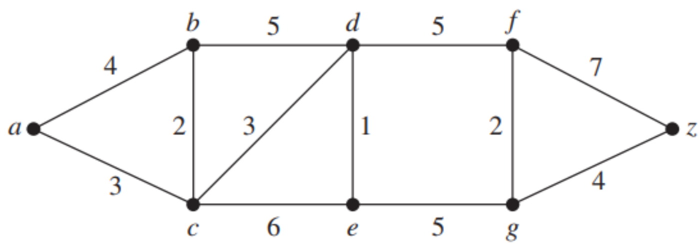
1.



2.



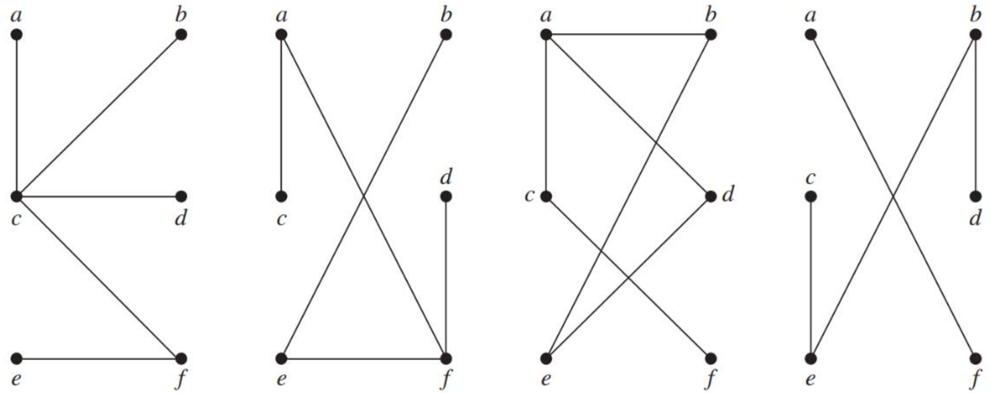
3.



TREE

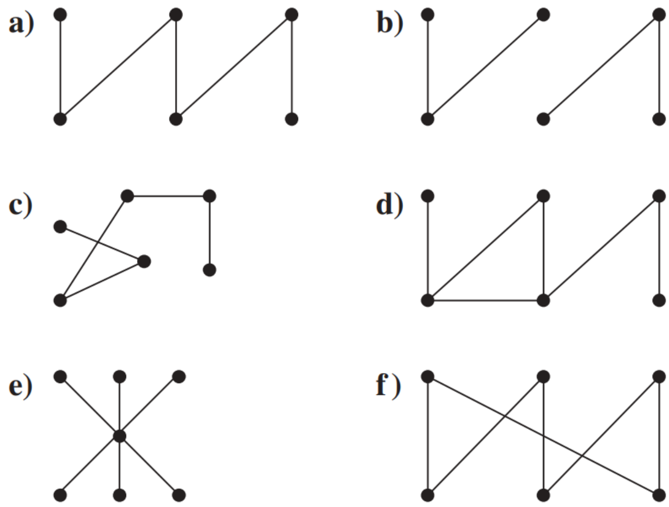
Introduction to Trees

1. Examples of trees and graphs that are not trees.

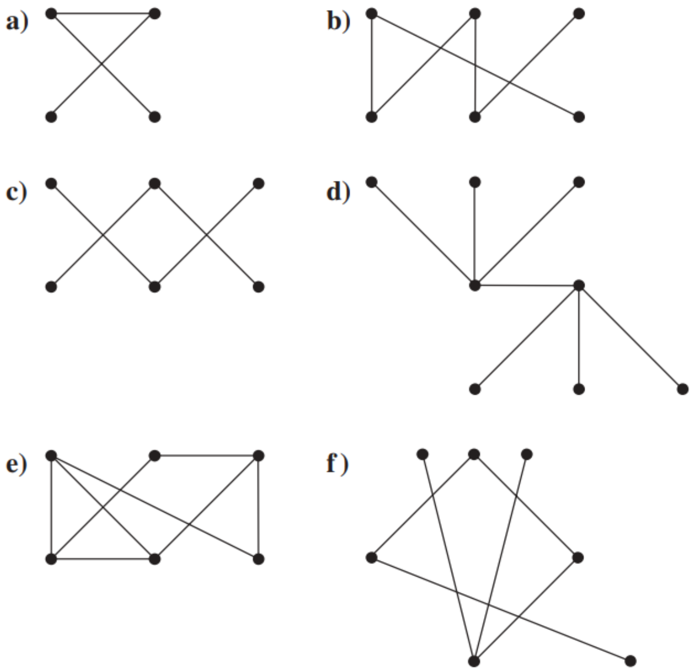


G_1 G_2 G_3 G_4

2. Which of these graphs are trees?

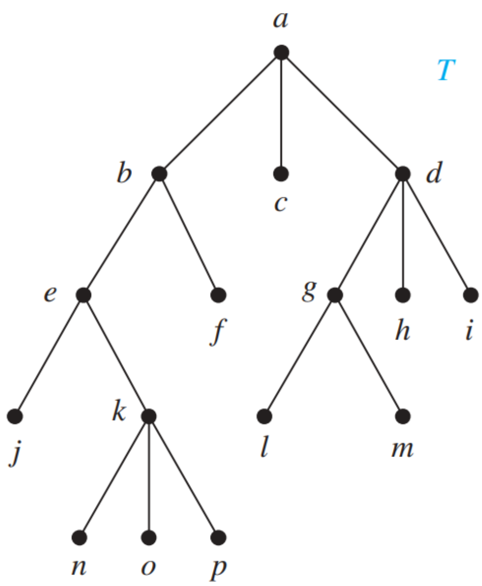


3. Which of these graphs are trees?

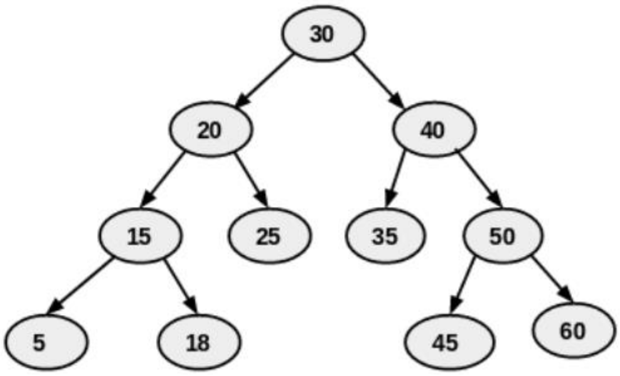


Tree Traversal

1. Find *in order, preorder*, and *pos t order* traversal of the following tree T. **(AU23)**

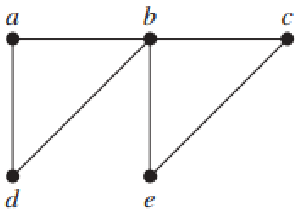


2. Find in order, preorder and post order traversal of the following tree **(SP23)**

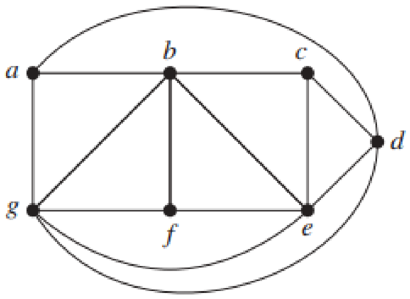


Spanning Trees

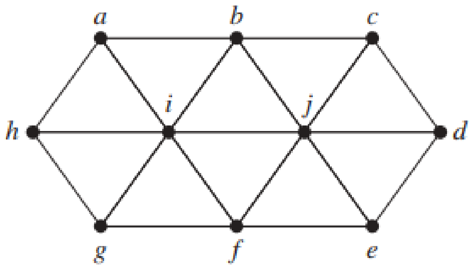
2.



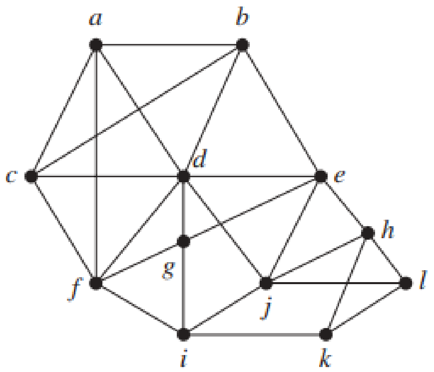
3.



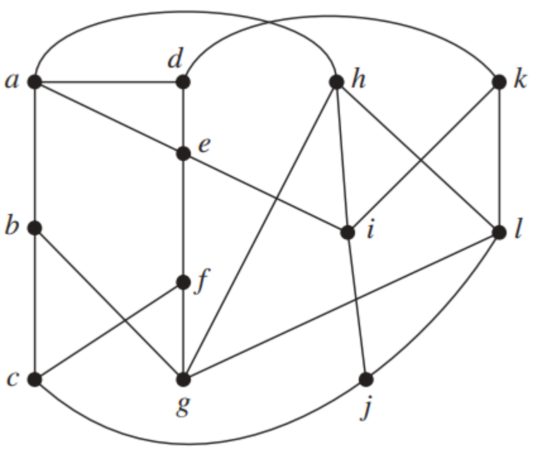
4.



5.



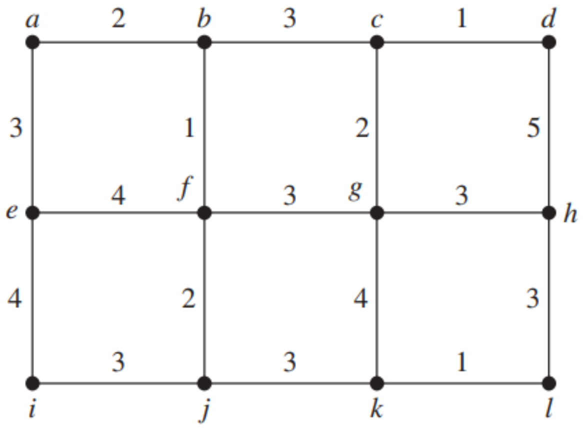
6.



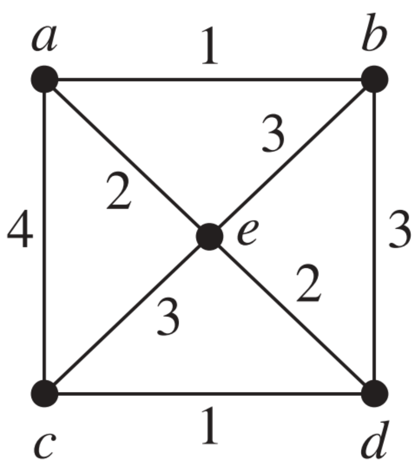
Minimum Spanning Trees

Prim's & Kruskal's algorithm to find a minimum spanning tree for the given weighted graph

1.



2.



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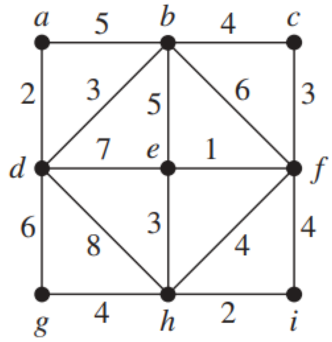

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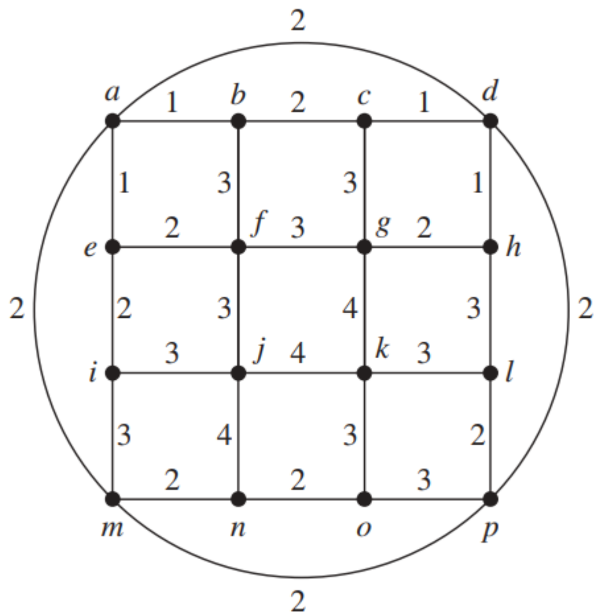

5.